ENERGY MARKET FUNDAMENTALS

Power Market Design

Market Liberalization, Grid Infrastructure, Trading Dynamics & The Merit Order Clearing Mechanism

CORE TOPICS

TSO/DSO Networks • Intraday Trading • Stack Dispatch

TARGET MODEL

Single Integrated EU Power Wholesale Price

REGULATORY EVOLUTION

Power Market Liberalization

Gas Market Framework

Power deregulation was modeled after gas market unbundling protocols.

Competition Challenges

State ownership and strategic generation assets created barrier-to-entry friction.

2000 Single Market Goal

Transitioned toward frictionless cross-border power flow and unified EU wholesale pricing.



GRID ARCHITECTURE

Infrastructure & Network Operations

TSO
Transmission System Operator

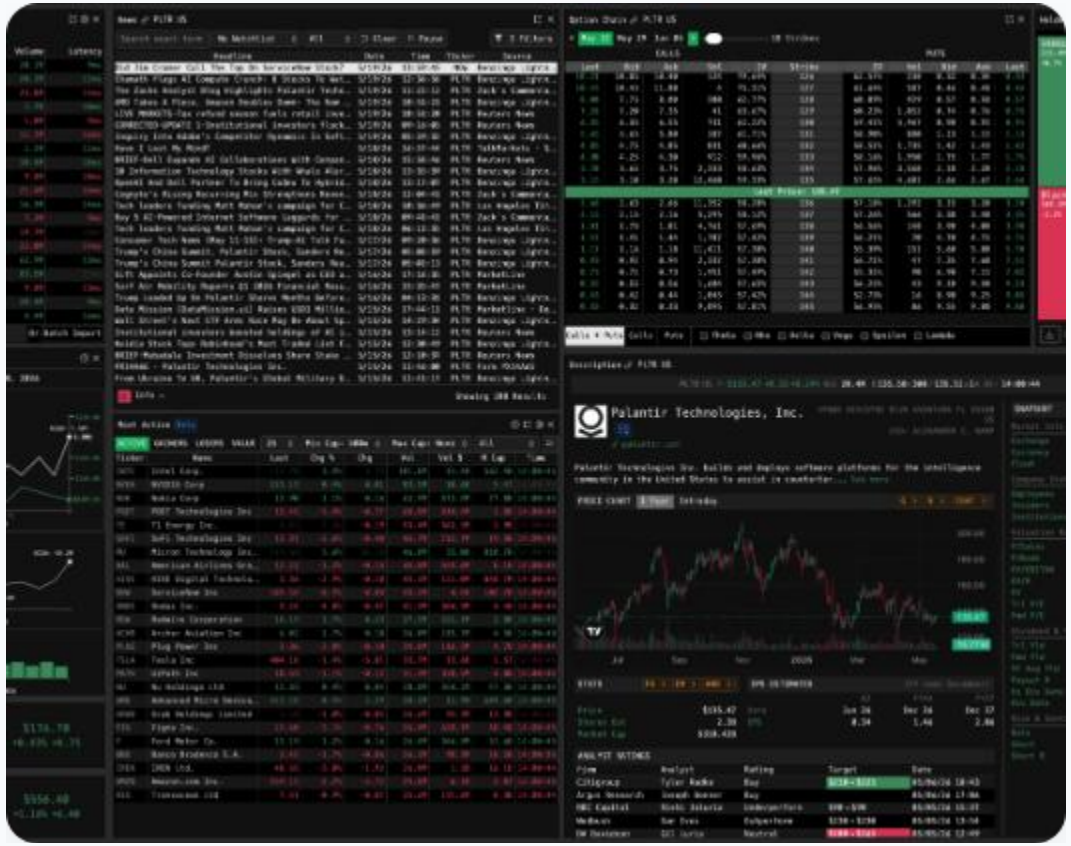
- **High Voltage Grid: Operates high-tension transmission lines and cross-border interconnectors.**
- **Wholesale Trading Hub: Acts as virtual trading hub where wholesale price clearing occurs.**

DSO
Distribution System Operator

- **Low Voltage Grid: Manages distribution lines to deliver power directly to end consumers.**
- **Tariff & Access Fees: Network usage, entry, and exit charges apply at every grid boundary.**

MARKET MECHANICS

Traded Products & Time Granularity



Non-Storability Impact

Forces intraday trading with 15-minute granularity, negative prices, and bottleneck congestion.

Intraday Physical Markets

Real-time auctions balance physical delivery. Benchmark metric: 1 MW powers ~1,000 households.

Financial Futures & Derivatives

Fixed-for-floating contracts with hourly fixing settlements hedge price volatility across hydro cycles.

ECONOMIC FOUNDATIONS

Supply, Demand & The Stack

Inelastic Demand

Power consumption remains virtually constant regardless of short-term price fluctuations.

The Merit Order Stack

Generators are ranked on the x-axis strictly from lowest to highest marginal operating cost.

Marginal Pricing

The clearing price is determined exactly where the vertical demand line intersects the generator supply stack.

MARKET CLEARING ILLUSTRATION

Illustrative Merit Order Stack

Market Price: €120 / MWh



1. Merit Order Rule: Generators dispatched strictly by lowest marginal cost.

2. Uniform Pricing: All dispatched units receive the €120 marginal clearing price.

3. Merit Shift: Renewables displace thermal assets, pushing stack price left.

MARKET STRUCTURE COMPARISON

Settlement Mechanisms

Pay-as-Cleared

EU Standard

- Every dispatched generator receives the identical marginal clearing price.
- Encourages generators to bid true marginal costs without gaming risk.

Pay-as-Bid

Alternative

- Each dispatched generator is paid its exact individual submitted offer bid.
- Increases strategic forecasting complexity as generators attempt to guess clearing limits.